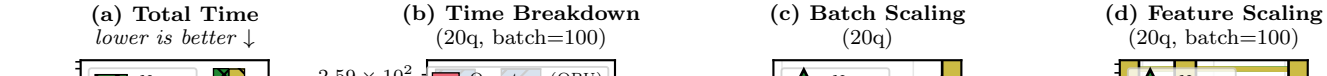
(a) Total Time
lower is better ↓
Total Time [s]
10^1
10^0
Naive
Batch
qTPU
20q
50q
100q
Circuit Size [qubits]
(b) Time Breakdown
(20q, batch=100)
Time [s]
2.59×10^2
2.58×10^2
2.57×10^2
2.56×10^2
2.55×10^2
2.54×10^2
2.53×10^2
Quantum (QPU)
Prep (CPU)
Classical
Naive Batch qTPU
Approach
(c) Batch Scaling
(20q)
Total Time [s]
10^1
10^0
Naive
Batch
qTPU
10^1
10^2
Batch Size
(d) Feature Scaling
(20q, batch=100)
Total Time [s]
10^0
Naive
Batch
qTPU
2.5
5.0
7.5
Feature Dimension